

T-tests and ANOVAs

Hypothesis Testing: Using t-tests and ANOVAs

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More Than Just “Eye-balling” It

Inferential statistics helps us to test hypotheses in precise ways.

Sometimes patterns are difficult to examine on our own, so we rely on certain statistical tools to clarify them.

Let's say 15 students took an exam. They did pretty well the first time (Pre-Score) but the professor thought they could do better if they tried again. Every student took the retake (Post-Score) knowing that they could either do better, the same, or worse on that version...

Here's Some Pretty Sus Data...

Name	Pre- Score	Post- Score
Eunice	81	85
Jerry	98	88
Clifford	81	88
Edith	99	99
Alice	87	91
Agatha	84	91
Cyrus	84	85
Jane	85	89
Chester	85	84
Howard	80	89
Janice	97	84
Abraham	86	98
Ambrose	86	95
Debra	92	98
Jethro	91	91

Did the class as a whole do better, worse, or the same?

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Better, right?

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Better, right?

But, wait, is there a *significant* difference in scores between the pre-test and the post-tests?

This Is Where t-tests Come In

T-tests are useful when we have both of the following:

- Continuous/Interval DV
- Dichotomous IV

In our pre-/post-test example, that's what we have! A continuous/interval variable (i.e., score) that we want to explain using a dichotomous variable (i.e., pre- or post-).

A t-test allows us to examine whether or not the means are significantly different across the two situations.

The Math (Just for Funsies)

Don't sweat the math steps to follow; they just illustrate what's going on with t-tests in the background.

$$t = \frac{\bar{x} - \mu}{se_m} \quad (1)$$

The t statistic is equal to the sample mean ($\bar{x}$) minus the mean we want to compare it to (μ), over the standard error of the mean (se_m).

$$se_m = \frac{s}{\sqrt{n}} \quad (2)$$

The standard error of the mean is equal to the sample standard deviation (s), divided by the square root of the sample size (n).

Some Subtraction and Averages

Name	Pre- Score	Post- Score	Difference
Eunice	81	85	4
Jerry	98	88	-10
Clifford	81	88	7
Edith	99	99	0
Alice	87	91	4
Agatha	84	91	7
Cyrus	84	85	1
Jane	85	89	4
Chester	85	84	-1
Howard	80	89	9
Janice	97	84	-13
Abraham	86	98	12
Ambrose	86	95	9
Debra	92	98	6
Jethro	91	91	0
AVERAGE	87.73	90.33	2.6

Steps to examining significance

1. Find our n and s : $n = 15$ and $s = 6.85$ (there are 15 students, and we'll just treat the standard deviation as given for now).
2. Calculate the t-statistic
3. Find our degrees of freedom (number of obs minus 1)
4. Compare the t-statistic to the critical value of t^*

First we need to calculate our standard error or the mean, then we can get our t-statistic.

$$se_m = \frac{s}{\sqrt{n}}$$

$$se_m = \frac{6.85}{\sqrt{15}} = 1.77$$

$$t = \frac{\bar{x} - \mu}{se_m}$$

$$t = \frac{90.33 - 87.73}{1.77} = 1.47$$

Finding the Critical Value

T-tests are called that because they rely on the Student's t-distribution.

- Degrees of freedom is $n - 1$ (14 for us) and we want a two-tailed significance level of 0.05 (remember 95% confidence and $p < 0.05$!)

TABLE 3—Percentage Points of Student's t Distribution

Degrees of Freedom			Level of Significance for a One-Tailed Test						
	.25	.20	.15	.10	.05	.025	.01	.005	.0005
(n-1)			Level of Significance for a Two-Tailed Test						
	.50	.40	.30	.20	.10	.05	.02	.01	.001
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	636.619
2	.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.598
3	.765	.978	1.250	1.638	2.353	3.182	4.541	5.841	12.924
4	.741	.941	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	.727	.920	1.156	1.476	2.015	2.571	3.365	4.032	6.869
6	.718	.906	1.134	1.440	1.943	2.447	3.143	3.707	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.998	3.499	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.896	3.355	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.821	3.250	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.764	3.169	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.718	3.106	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.681	3.055	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.650	3.012	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.624	2.977	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.602	2.947	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.583	2.921	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.567	2.898	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.552	2.878	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850

What is our critical value of t^* ?

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2.145

Did Students' Scores Significantly Improve?

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Nope. $t < t^*$

To see whether or not the scores improved *significantly*, we have to compare our calculated t-statistic to the critical value of t^* and ensure that it's larger.

Because our t-statistic (1.47) is less than our critical t^* value (2.145)...

- we **fail to reject** the null hypothesis.
- we can conclude that the difference in pre- and post-test scores is **not statistically significant**, even though we haven't calculated the p-value (which is, ugh, too much work).

But Wait, There's More!

What if you have a continuous/interval DV but you have more than 2 categories in the IV?

This Is Where ANOVAs Come In

ANOVAs are useful when we have both of the following:

- Continuous/Interval DV
- Ordinal or Nominal IV

ANOVA is short for “ANalysis Of VAriance”

- We're still comparing means, though...
- The “variance” part of ANOVAs is important because the test relies on an assumption: that the variances between groups are equal (or, not significantly different).
- Instead of a t-distribution, an ANOVA relies on an F-distribution (don't worry about it)

An ANOVA allows us to examine whether or not the mean of our DV is significantly different across the 3 or more categories of the IV.